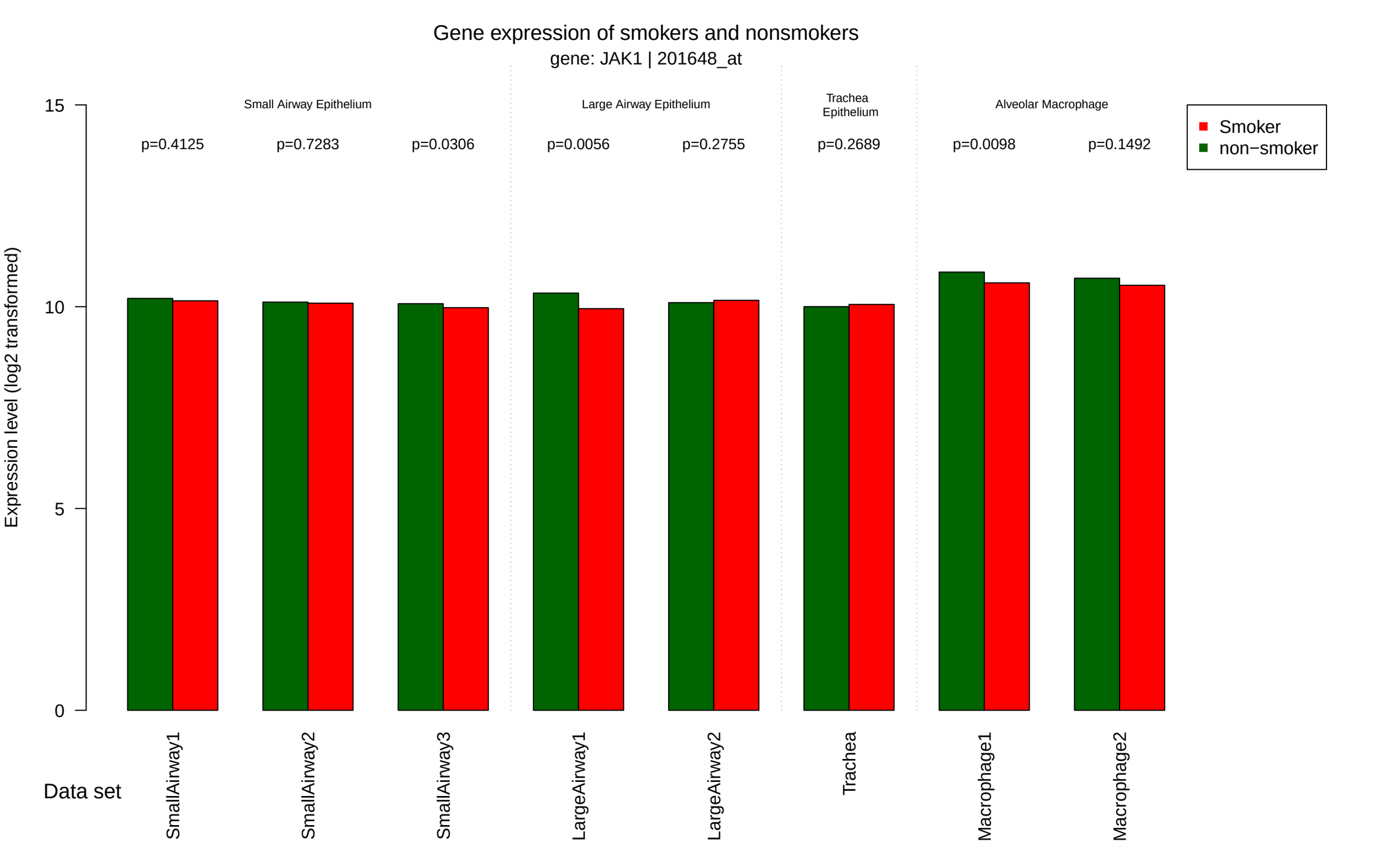
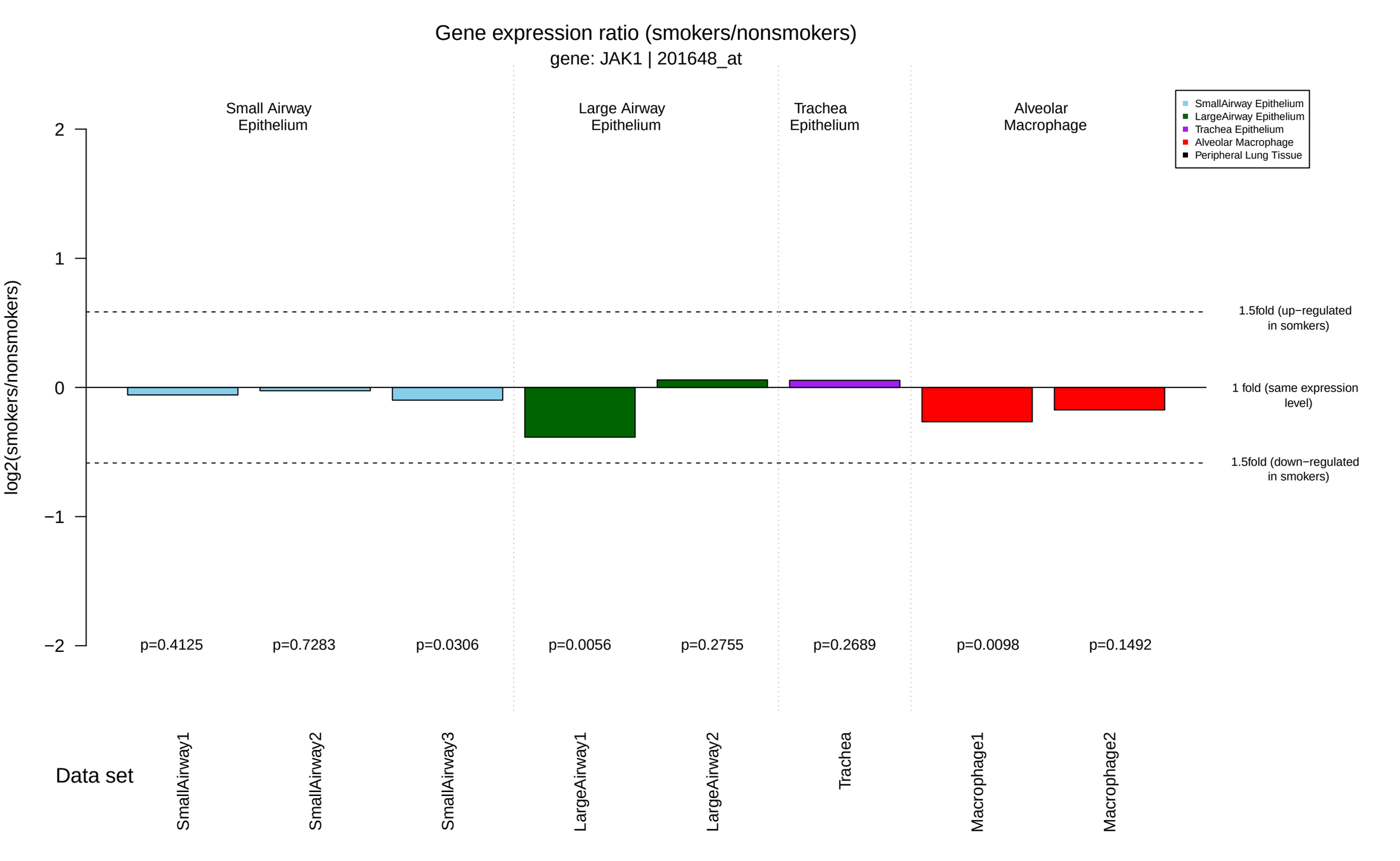
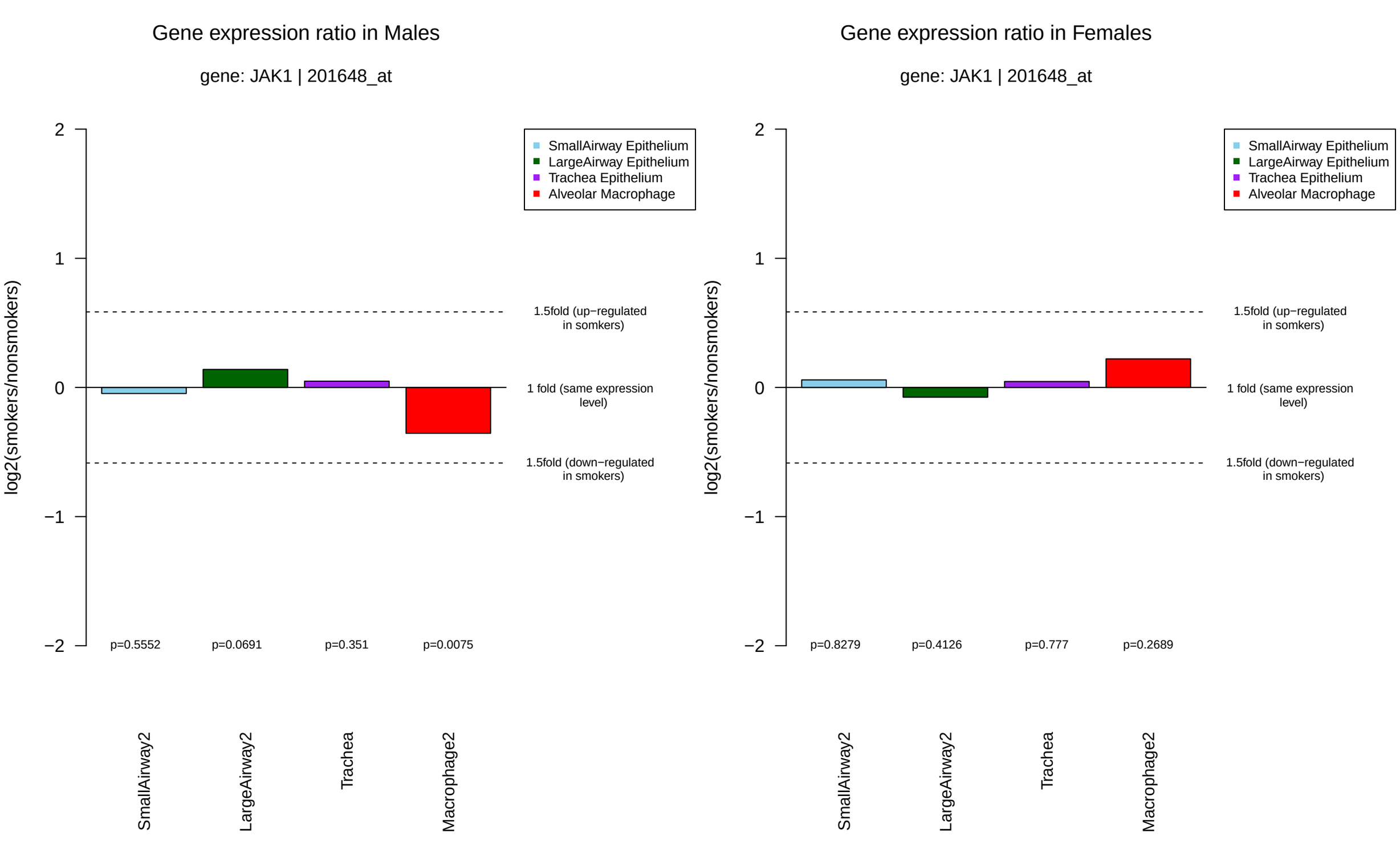
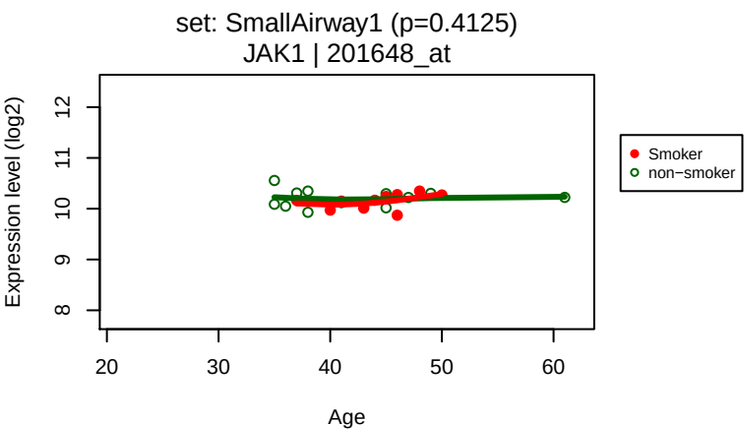
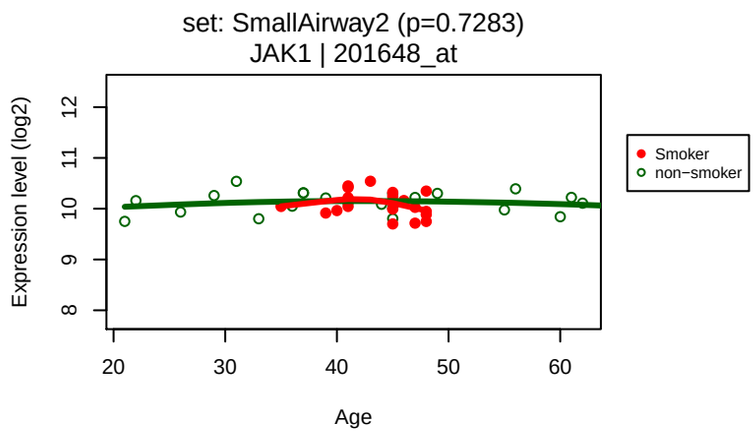
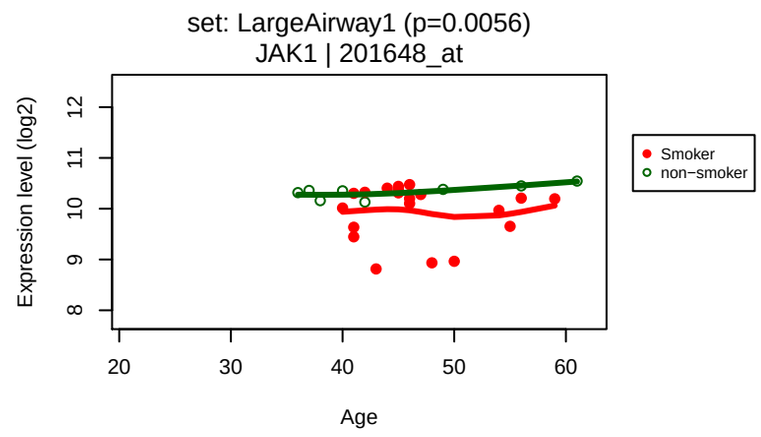

Gene expression of smokers and nonsmokers
gene: JAK1 | 201648_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4125
p=0.7283
p=0.0306
p=0.0056
p=0.2755
p=0.2689
p=0.0098
p=0.1492
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

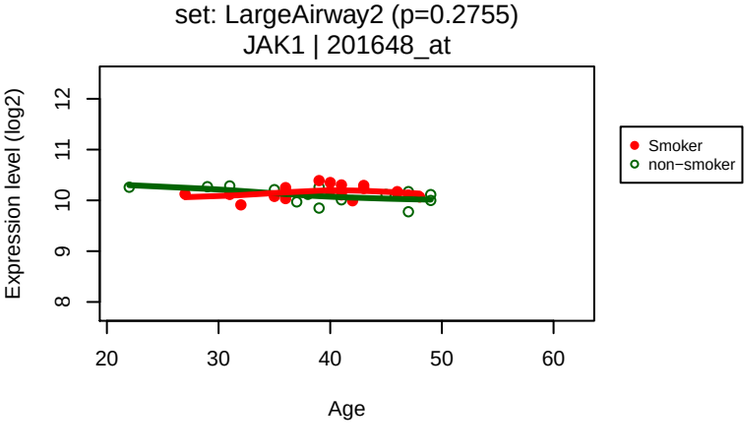

Gene expression ratio (smokers/nonsmokers)
gene: JAK1 | 201648_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4125
p=0.7283
p=0.0306
p=0.0056
p=0.2755
p=0.2689
p=0.0098
p=0.1492
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

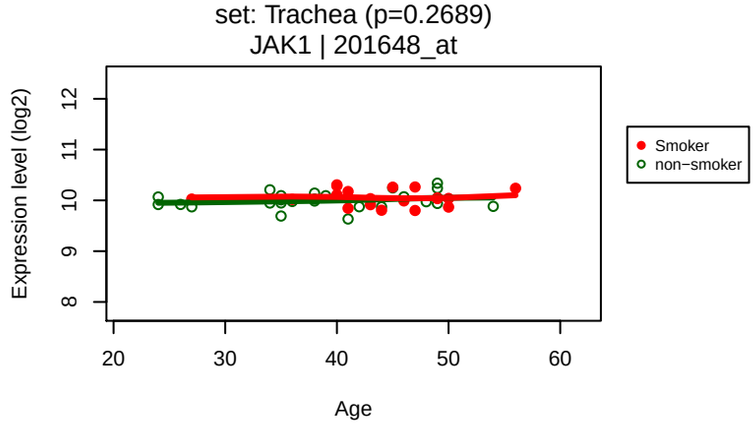

Gene expression ratio in Males
gene: JAK1 | 201648_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5552
p=0.0691
p=0.351
p=0.0075
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK1 | 201648_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8279
p=0.4126
p=0.777
p=0.2689
SmallAirway2
LargeAirway2
Trachea
Macrophage2

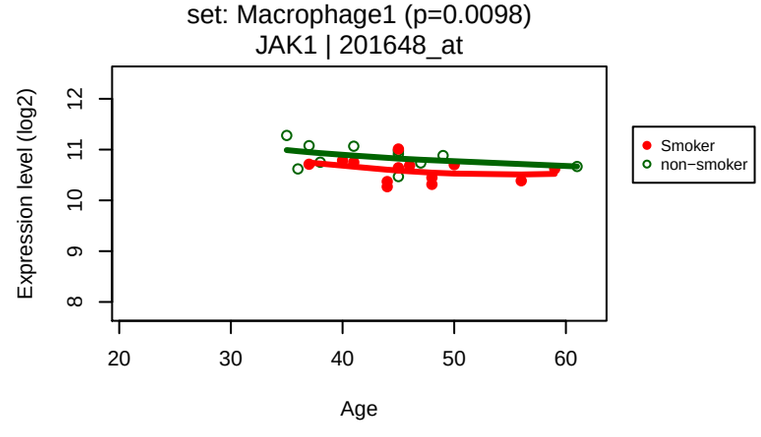

set: SmallAirway1 (p=0.4125)
JAK1 | 201648_at
Expression level (log2)
Age
Smoker
non−smoker

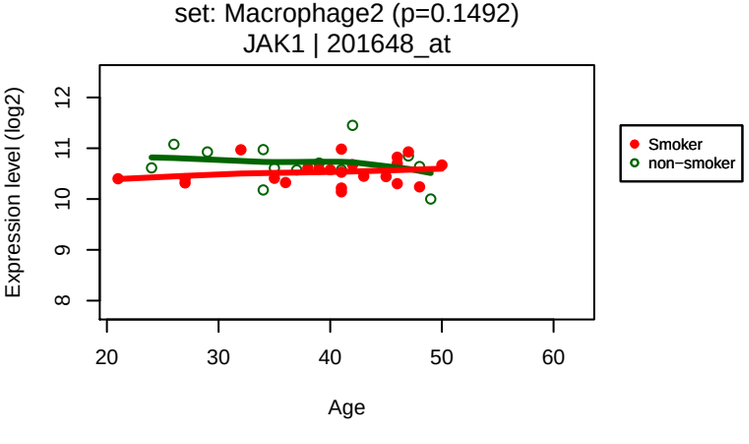

set: SmallAirway2 (p=0.7283)
JAK1 | 201648_at
Expression level (log2)
Age
Smoker
non−smoker

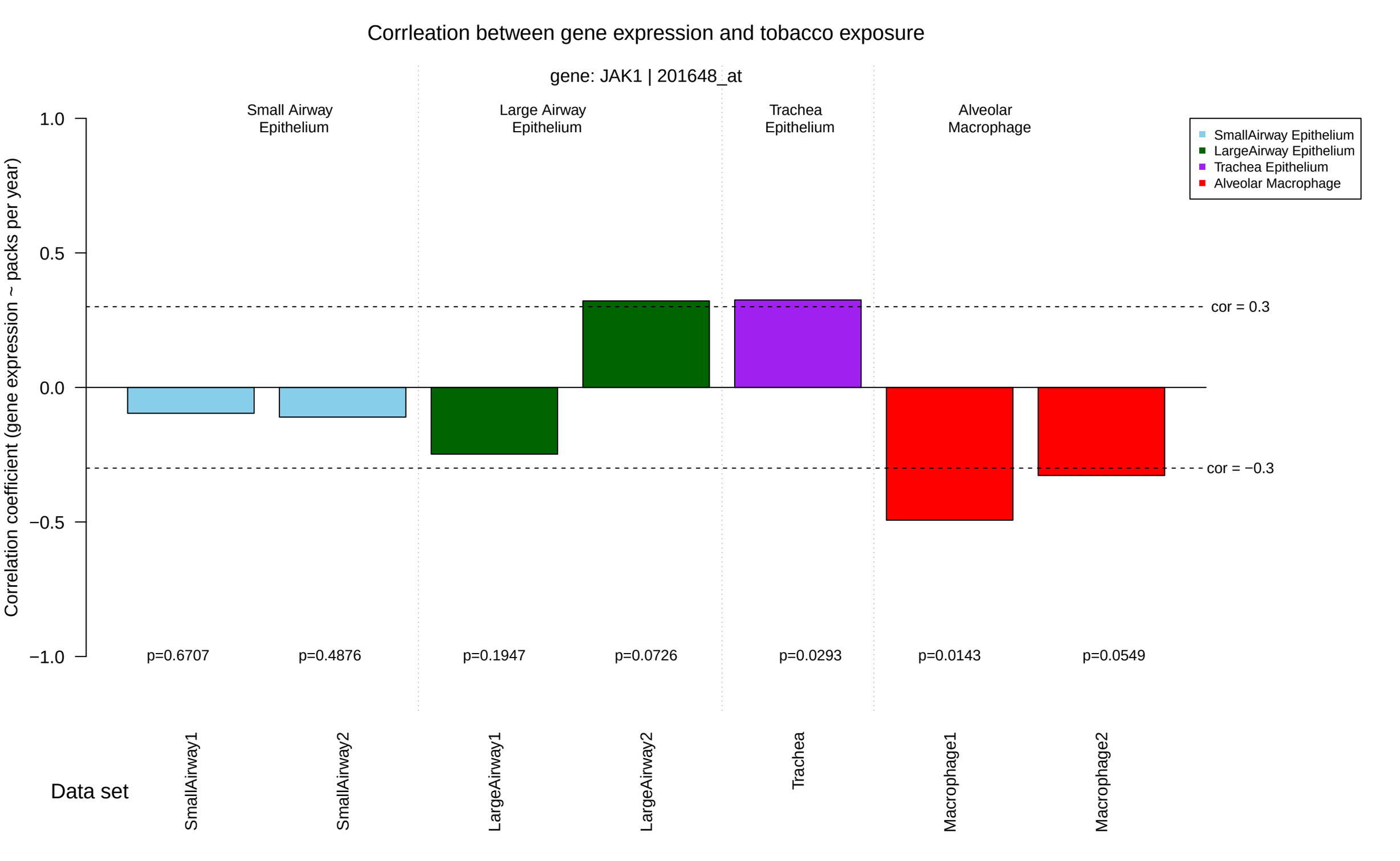

set: LargeAirway1 (p=0.0056)
JAK1 | 201648_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2755)
JAK1 | 201648_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2689)
JAK1 | 201648_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0098)
JAK1 | 201648_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1492)
JAK1 | 201648_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK1 | 201648_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6707
p=0.4876
p=0.1947
p=0.0726
p=0.0293
p=0.0143
p=0.0549
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2